

Paper Reference(s)

Examiner's use only

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Team Leader's use only

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When a calculator is used, the answer should be given to an appropriate degree of accuracy.

There are 16 pages in this question paper. Any blank pages are indicated.

You must show sufficient working to make your methods clear to the Examiner. Answers without working may gain no credit.

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1.

(4)

2. Two particles A and B have mass 0.4 kg and 0.3 kg respectively. They are moving in opposite directions on a smooth horizontal table and collide directly. Immediately before the collision, the speed of A is 6 m s^{-1} and the speed of B is 2 m s^{-1} . As a result of the collision, the direction of motion of B is reversed and its speed immediately after the collision is 3 m s^{-1} . Find

- (4)

- (3)

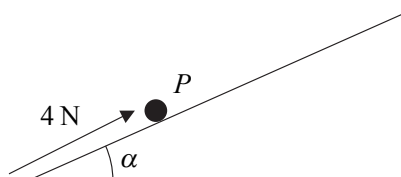




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4.

Figure 2



A particle P of mass 0.5 kg is on a rough plane inclined at an angle α to the horizontal, where $\tan \alpha = \frac{3}{4}$. The particle is held at rest on the plane by the action of a force of magnitude 4 N acting up the plane in a direction parallel to a line of greatest slope of the plane, as shown in Figure 2. The particle is on the point of slipping up the plane.

- (a) Find the coefficient of friction between P and the plane.

(7)

The force of magnitude 4 N is removed.

- (b) Find the acceleration of P down the plane.

(4)



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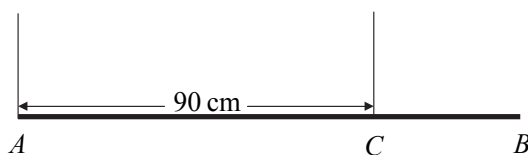
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5.

Figure 3



A steel girder AB has weight 210 N . It is held in equilibrium in a horizontal position by two vertical cables. One cable is attached to the end A . The other cable is attached to the point C on the girder, where $AC = 90\text{ cm}$, as shown in Figure 3. The girder is modelled as a uniform rod, and the cables as light inextensible strings.

Given that the tension in the cable at C is twice the tension in the cable at A , find

- (a) the tension in the cable at A , **(2)**
- (b) show that $AB = 120$ cm. **(4)**

A small load of weight W newtons is attached to the girder at B . The load is modelled as a particle. The girder remains in equilibrium in a horizontal position. The tension in the cable at C is now three times the tension in the cable at A .

- (c) Find the value of W . (7)



Q5

(Total 13 marks)





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7. [In this question the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are due east and north respectively.]

A ship S is moving with constant velocity $(-2.5\mathbf{i} + 6\mathbf{j}) \text{ km h}^{-1}$. At time 1200, the position vector of S relative to a fixed origin O is $(16\mathbf{i} + 5\mathbf{j}) \text{ km}$. Find

- (a) the speed of S ,
- (2)**

- (b) the bearing on which S is moving. (2)

The ship is heading directly towards a submerged rock R . A radar tracking station calculates that, if S continues on the same course with the same speed, it will hit R at the time 1500.

- (c) Find the position vector of R . (2)

The tracking station warns the ship's captain of the situation. The captain maintains S on its course with the same speed until the time is 1400. He then changes course so that S moves due north at a constant speed of 5 km h^{-1} . Assuming that S continues to move with this new constant velocity, find

- (d) an expression for the position vector of the ship t hours after 1400,

- (e) the time when S will be due east of R ,

- (f) the distance of S from R at the time 1600. (3)



Question 7 continued

Q7

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END

